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Prompt Engineering with OpenAI

```
"A good argument is never lost."
Publilius Syrus
```

The following examples provide an introduction to prompt engineering using OpenAI's API. Various prompt engineering techniques, beginning with basic prompt examples that explore adjusting parameters like temperature to influence response style. It then progresses into more advanced prompting techniques, including few-shot and chain-of-thought (CoT) examples, designed to enhance response quality for different types of tasks.

Install libraries: `pip install openai python-dotenv langchain langchain-community` ↩

Example `env` file named `teachingkey.env` has the following line:
`OPENAI_API_KEY=<your secret API key from OpenAI>`

The API key file is acquired from ✳ **OpenAI** webpage.

The following examples might need an account and a small positive account balance ($). Run the Cells slowly to avoid hitting rate limits.

The first two examples are basic prompts and changing the prompt/response temperature. Next are various examples, including few-shot prompts, CoT, zero-shot CoT, etc.

In [1]:
```
%matplotlib inline
import matplotlib.pyplot as plt
plt.rcParams['figure.dpi'] = 72
from IPython.display import Markdown
import numpy as np
import os

import openai
import IPython
from langchain.llms import OpenAI
from dotenv import load_dotenv
```

In [2]:
```
load_dotenv('../teachingkey.env')

# API configuration
Client = openai.OpenAI(api_key=os.environ.get('OPENAI_API_KEY', '<your Ope
```

In [3]:
```
def set_openai_params(model='gpt-3.5-turbo', temp=0.7, max_tokens=256, top
    openai_params = {
        'model': model,
        'temperature': temp,
        'max_tokens': max_tokens,
        'top_p': top_p,
        'frequency_penalty': frequency_penalty,
        'presence_penalty': presence_penalty
    }
    return openai_params

def get_completion(params, messages):
    # GET completion from OpenAI API
    response = Client.chat.completions.create(
        model = params['model'],
        messages = messages,
        temperature = params['temperature'],
        max_tokens = params['max_tokens'],
        top_p = params['top_p'],
        frequency_penalty = params['frequency_penalty'],
        presence_penalty = params['presence_penalty'],
    )
    return response
```

In [4]:
```
# a basic example
params = set_openai_params()
prompt = 'The sky is'

messages = [{"role":"user", "content":prompt}]
response = get_completion(params, messages)
Markdown(response.choices[0].message.content)
```

Out[4]: blue and clear, with fluffy white clouds scattered across it. The sun is shining brightly, warming everything it touches. It's a beautiful day to be outside and enjoy the natural beauty of the world.

In [5]:
```
# change temperature
params = set_openai_params(temp=0)

response = get_completion(params, messages)
Markdown(response.choices[0].message.content)
```

Out[5]: blue and clear, with fluffy white clouds scattered across the horizon. The sun is shining brightly, casting a warm glow over everything below. It's a beautiful day to be outside and enjoy the beauty of nature.

In [6]:
```
# text summarization
params = set_openai_params(temp=0.7)
prompt = """Antibiotics are a type of medication used to treat bacterial i
They work by either killing the bacteria or preventing them from reproduci
Antibiotics are usually taken orally in the form of pills, capsules, or li
They are not effective against viral infections, and using them inappropri

Explain the above in one sentence:"""

messages = [{"role":"user", "content":prompt}]

response = get_completion(params, messages)
Markdown(response.choices[0].message.content)
```

Out[6]: Antibiotics are medications that treat bacterial infections by either killing the bacteria or preventing their reproduction, but they are not effective against viral infections and should be used properly to prevent antibiotic resistance.

In [7]:
```
# question answering
prompt = """Answer the question based on the context below. Keep the answe
Respond "Unsure about answer" if not sure about the answer.

Context: Teplizumab traces its roots to a New Jersey drug company called O
There, scientists generated an early version of the antibody, dubbed OKT3.
Originally sourced from mice, the molecule was able to bind to the surface
In 1986, it was approved to help prevent organ rejection after kidney tran

Question: What was OKT3 originally sourced from?

Answer:"""

messages = [{"role":"user", "content":prompt}]

response = get_completion(params, messages)
Markdown(response.choices[0].message.content)
```

Out[7]: Mice

In [8]:
```
# text classification
prompt = """Classify the text into neutral, negative or positive.

Text: I think the food was okay.

Sentiment:"""

messages = [{"role":"user", "content":prompt}]

response = get_completion(params, messages)
Markdown(response.choices[0].message.content)
```

Out[8]: Neutral

In [9]:
```
# role playing
prompt = """The following is a conversation with an AI research assistant.

Human: Hello, who are you?
AI: Greeting! I am an AI research assistant. How can I help you today?
Human: Can you tell me about the creation of blackholes?
AI:"""

messages = [{"role":"user", "content":prompt}]

response = get_completion(params, messages)
Markdown(response.choices[0].message.content)
```

Out[9]: Certainly. Black holes are formed when massive stars run out of nuclear fuel and collapse under their own gravity. This causes the star's core to shrink to a point of infinite density known as a singularity, surrounded by an event horizon from which nothing, not even light, can escape. This results in a region of spacetime with extremely strong gravitational forces, known as a black hole.

In [10]:
```
# code generation
prompt = """
"\"\"\"\nTable departments, columns = [DepartmentId, DepartmentName]\nTabl
columns = [DepartmentId, StudentId, StudentName]\nCreate a MySQL query for
"""

messages = [{"role":"user", "content":prompt}]

response = get_completion(params, messages)
Markdown(response.choices[0].message.content)
```

Out[10]: SELECT s.StudentId, s.StudentName FROM students s JOIN departments d ON s.DepartmentId = d.DepartmentId WHERE d.DepartmentName = 'Computer Science';

In [11]:
```
# reasoning
prompt = """The odd numbers in this group add up to an even number: 15, 32

Solve by breaking the problem into steps. First, identify the odd numbers,

messages = [{"role":"user", "content":prompt}]

response = get_completion(params, messages)
Markdown(response.choices[0].message.content)
```

Out[11]: Odd numbers in the group: 15, 5, 13, 7, 1

15 + 5 + 13 + 7 + 1 = 41

41 is an odd number.

In [12]:
```
# few shot prompts
prompt = """The odd numbers in this group add up to an even number: 4, 8,
A: The answer is False.

The odd numbers in this group add up to an even number: 17,  10, 19, 4, 8,
A: The answer is True.

The odd numbers in this group add up to an even number: 16,  11, 14, 4, 8,
A: The answer is True.

The odd numbers in this group add up to an even number: 17,  9, 10, 12, 13
A: The answer is False.

The odd numbers in this group add up to an even number: 15, 32, 5, 13, 82,
A:"""

messages = [{"role":"user", "content":prompt}]

response = get_completion(params, messages)
Markdown(response.choices[0].message.content)
```

Out[12]: The answer is False.

In [13]:
```
# chain-of-thought
prompt = """The odd numbers in this group add up to an even number: 4, 8,
A: Adding all the odd numbers (9, 15, 1) gives 25. The answer is False.

The odd numbers in this group add up to an even number: 15, 32, 5, 13, 82,
A:"""

messages = [{"role":"user", "content":prompt}]

response = get_completion(params, messages)
Markdown(response.choices[0].message.content)
```

Out[13]: Adding all the odd numbers (15, 5, 13, 7, 1) gives 41. The answer is False.

In [14]:
```
# zero shot CoT
prompt = """I went to the market and bought 10 apples.
I gave 2 apples to the neighbor and 2 to the repairman.
I then went and bought 5 more apples and ate 1.
How many apples did I remain with?

Let's think step by step."""

messages = [{"role":"user", "content":prompt}]

response = get_completion(params, messages)
Markdown(response.choices[0].message.content)
```

Out[14]: Initially, you bought 10 apples. After giving 2 apples to the neighbor and 2 to the repairman, you had 10 - 2 - 2 = 6 apples left.

Then you bought 5 more apples, so you had 6 + 5 = 11 apples.

After eating 1 apple, you had 11 - 1 = 10 apples remaining.

So, you remained with 10 apples.

In [15]:
```
# self-consistency
question = "What are some good activities to relax after a long day?"

# generate multiple responses for self-consistency
responses = []
for _ in range(5):  # 5 responses for consistency
    messages = [{"role": "user", "content": question}]
    response = get_completion(params, messages)
    responses.append(response.choices[0].message.content.strip())

for idx, resp in enumerate(responses, start=1):
    print(f"Response {idx}:\n {resp}")

# select the most frequent answer
from collections import Counter
most_common_answer = Counter(responses).most_common(1)[0][0]

print("\n  ** The most consistent answer:\n", most_common_answer)
```

Response 1:
1. Take a warm bath or shower
2. Practice deep breathing or meditation
3. Listen to calming music or nature sounds
4. Do some gentle stretching or yoga
5. Read a book or watch a movie
6. Spend time in nature, such as going for a walk or sitting outside
7. Write in a journal or do some creative writing
8. Drink a cup of herbal tea
9. Engage in a hobby or craft that you enjoy
10. Spend time with a pet or loved one
11. Practice mindfulness or visualization techniques
12. Treat yourself to a soothing massage or self-care routine.

Response 2:
1. Take a warm bath or shower
2. Practice deep breathing or meditation
3. Listen to calming music or nature sounds
4. Read a book or watch a movie
5. Go for a walk or do some gentle stretching exercises
6. Write in a journal or do some creative writing
7. Drink a cup of herbal tea or hot cocoa
8. Do some light gardening or engage in a hobby you enjoy
9. Spend time with loved ones or pets
10. Practice mindfulness or visualization techniques.

Response 3:
1. Take a warm bath or shower
2. Practice yoga or meditation
3. Read a book or listen to calming music
4. Go for a walk or do some light exercise
5. Journal or write down your thoughts and feelings
6. Watch a movie or TV show
7. Cook a nice meal or bake something
8. Have a cup of tea or hot chocolate
9. Spend time with loved ones or pets
10. Do a creative activity like drawing or crafting.

Response 4:
1. Take a warm bath or shower
2. Practice meditation or deep breathing exercises
3. Listen to calming music or nature sounds
4. Do some gentle stretching or yoga
5. Read a book or watch a movie
6. Spend time in nature or go for a walk
7. Engage in a creative hobby like drawing or painting
8. Write in a journal or practice gratitude
9. Enjoy a cup of tea or hot cocoa
10. Spend time with loved ones or pets.

Response 5:
1. Take a warm bath or shower
2. Practice deep breathing or meditation
3. Listen to calming music or nature sounds
4. Do some gentle yoga or stretching
5. Read a book or watch a favorite movie or TV show
6. Take a walk outside in nature
7. Write in a journal or do some creative writing
8. Cook a comforting meal or bake some treats
9. Have a cup of herbal tea or hot cocoa

```
10. Spend time with loved ones or pets.

  ** The most consistent answer:
 1. Take a warm bath or shower
2. Practice deep breathing or meditation
3. Listen to calming music or nature sounds
4. Do some gentle stretching or yoga
5. Read a book or watch a movie
6. Spend time in nature, such as going for a walk or sitting outside
7. Write in a journal or do some creative writing
8. Drink a cup of herbal tea
9. Engage in a hobby or craft that you enjoy
10. Spend time with a pet or loved one
11. Practice mindfulness or visualization techniques
12. Treat yourself to a soothing massage or self-care routine.
```

In [16]:

```python
# generated knowledge prompting
question = "Why do planets orbit the sun?"

# generated knowledge steps
knowledge_prompts = [
    "Explain the concept of gravitational force.",
    "Describe how gravity affects objects in space.",
    "Explain how an object's mass influences the gravitational force it ex
    "Describe how gravitational force affects planetary motion around the
]

# collect knowledge statements
knowledge_statements = []
for k_prompt in knowledge_prompts:
    messages = [{"role": "user", "content": k_prompt}]
    response = get_completion(params, messages)
    knowledge_statements.append(response.choices[0].message.content.strip(

# combine the knowledge statements to answer the main question
combined_knowledge = " ".join(knowledge_statements)
final_prompt = f"{combined_knowledge} Now, based on this information, why

# generate the final response using the accumulated knowledge
messages = [{"role": "user", "content": final_prompt}]
final_response = get_completion(params, messages)

print("\n  ** Accumulated Knowledge Statements:")
for idx, statement in enumerate(knowledge_statements, start=1):
    print(f"Statement {idx}: {statement}")

print("\n  ** Final Answer:", final_response.choices[0].message.content.st
```

** Accumulated Knowledge Statements:
Statement 1: Gravitational force is the force of attraction between two objects with mass. According to Newton's Law of Universal Gravitation, every object in the universe attracts every other object with a force that is directly proportional to the product of their masses and inversely proportional to the square of the distance between their centers. This means that the larger the mass of the objects and the closer they are to each other, the stronger the gravitational force between them.

Gravitational force is responsible for keeping planets in orbit around stars, moons in orbit around planets, and objects on the surface of the Earth. It is also the force that causes objects to fall towards the Earth when they are dropped. The strength of the gravitational force depends on the masses of the objects and the distance between them, and it is one of the fundamental forces of nature that govern the motion of objects in the universe.
Statement 2: Gravity is a fundamental force that affects all objects in space, including planets, stars, asteroids, and spacecraft. It is responsible for keeping planets in orbit around stars, moons in orbit around planets, and objects falling towards larger masses.

The force of gravity depends on the mass of the objects and the distance between them. The greater the mass of an object, the stronger its gravitational pull. For example, the larger the planet, the stronger its gravitational pull on objects near it.

In space, objects are constantly affected by the gravitational forces of nearby masses. This is why planets and moons stay in orbit around stars and planets, and why comets and asteroids are attracted to planets and other large bodies.

Gravity also affects the motion of objects in space. For example, the gravitational pull of a planet can cause a spacecraft to be pulled towards it, altering its trajectory. Gravity can also cause objects to collide with each other or be pulled apart if they get too close.

Overall, gravity plays a crucial role in the dynamics of objects in space, shaping the orbits of planets and moons, and influencing the movement of objects throughout the universe.
Statement 3: An object's mass directly influences the gravitational force it exerts on other objects. According to Newton's law of universal gravitation, the gravitational force between two objects is directly proportional to the product of their masses and inversely proportional to the square of the distance between their centers.

This means that the greater the mass of an object, the stronger the gravitational force it will exert on other objects. For example, the Earth has a much larger mass than a small rock, so it exerts a much stronger gravitational force on the rock. Similarly, a more massive object will experience a greater gravitational force when attracted towards another object.

In summary, the gravitational force exerted by an object is directly proportional to its mass. This principle is essential for understanding the behavior of objects in space and on Earth, as gravity plays a significant role in determining the motion and interactions of celestial bodies.
Statement 4: Gravitational force is the force of attraction between two objects with mass. In the case of planetary motion around the sun, the gravitati

onal force between the planet and the sun is what keeps the planet in orbit around the sun.

The gravitational force between the planet and the sun is what causes the planet to move in a curved path around the sun. This force acts as a centripetal force, pulling the planet towards the sun and keeping it in orbit.

The strength of the gravitational force between the planet and the sun is determined by the masses of the two objects and the distance between them. The greater the mass of the sun and the planet, the stronger the gravitational force between them. Similarly, the closer the planet is to the sun, the stronger the gravitational force.

As a result, the gravitational force between the planet and the sun causes the planet to move in an elliptical orbit around the sun. The planet will move faster when it is closer to the sun and slower when it is farther away, in accordance with Kepler's laws of planetary motion.

Overall, gravitational force plays a crucial role in determining the motion of planets around the sun, keeping them in orbit and maintaining the stability of the solar system.

** Final Answer: Planets orbit the sun because of the gravitational force between the planets and the sun. According to Newton's Law of Universal Gravitation, every object in the universe attracts every other object with a force that is directly proportional to the product of their masses and inversely proportional to the square of the distance between their centers. This means that the sun's massive gravitational pull keeps the planets in orbit around it.

The gravitational force between the sun and the planets acts as a centripetal force, pulling the planets towards the sun and keeping them in their elliptical orbits. The strength of this gravitational force is determined by the masses of the sun and the planets, as well as the distance between them. The closer a planet is to the sun, the stronger the gravitational force, which causes the planet to move faster in its orbit.

In summary, planets orbit the sun because of the gravitational force between them. This force keeps the planets in their orbits and determines their motion around the sun, following the laws of planetary motion established by Kepler.

In [17]:
```
%%html
<style>
    table {margin-left: 0 !important;}
    p {font-family: verdana;}
    li {font-family: verdana;}
    div {font-size: 10pt;}
</style>
<!-- Display markdown tables left oriented in this notebook. -->
```